Inter-City Parcel Routing Through a Hub Network

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A system design framework for optimal middle-mile routing, load consolidation, and dispatch decisions (Delhi → Mumbai case)

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1. Reframing the Problem

This is a well-studied class of problem in freight/logistics operations research called **Service Network Design (SND)**, combined with a **time-dependent shortest path problem** for routing each individual shipment through the resulting network. It splits into two distinct layers that are easy to conflate but should be solved separately:

1. **Strategic layer (planned periodically, e.g. weekly/monthly):** Which hub-to-hub "lanes" does the company actually operate trucks on, at what frequency, and with what capacity? Not every warehouse connects directly to every other — there's a designed lane network, similar to how an airline decides its route map.
2. **Operational layer (solved per-shipment, in real time):** Given that fixed(ish) lane network with its schedules and capacities, what's the fastest way to get *this specific parcel* from Delhi to Mumbai right now?

The key mental shift: **routing a parcel through this network is not a "find the geographically shortest path between warehouses" problem — it's much closer to booking a flight itinerary.** You don't just pick the nearest airport as your next stop; you pick based on which flights are actually scheduled, when they leave, whether there's a seat, and what the total journey time is including layovers. Your intuition about "figuring the next best warehouse" is exactly this — and the formal tool for it is a **time-expanded network**, covered in Section 5.

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2. Data You Need

Category	Specifics
Hub network graph	Which warehouse pairs have direct scheduled trucking lanes (not a full mesh — most hub pairs are *not* directly connected), distance and typical transit time per lane
Departure schedules & frequency	For each lane: how many trucks per day, at what times, historical on-time performance
Truck capacity & current load	Volume/weight capacity per truck, real-time fill level for trucks not yet dispatched
Hub processing data	Sorting/dwell time distribution at each hub — how long a parcel typically sits between arrival and being loaded onto the next leg
Parcel volume history/forecast	Historical and forecast parcel counts per lane per time-of-day — needed to decide whether to wait for consolidation or dispatch now

Real-time truck tracking	Live GPS location of every truck currently in transit, to detect delays and trigger re-planning
Cost data	Per-km transportation cost, per-hub handling cost, driver cost, and — critically — the cost of delay (SLA penalty, customer dissatisfaction) so the two can be traded off explicitly
Driver regulation data	Maximum continuous driving hours, mandated rest periods, relay/driver-change hub locations
Road/route condition data	Highway conditions, known bottlenecks, seasonal risk (e.g., monsoon-prone stretches), toll plaza delays on the specific corridor
SLA/priority tier definitions	Express vs. standard vs. economy shipments, and whether each has a distinct routing/dispatch policy
Multi-modal options	Availability, cost, and transit time for rail freight or air cargo alternatives, if used alongside road

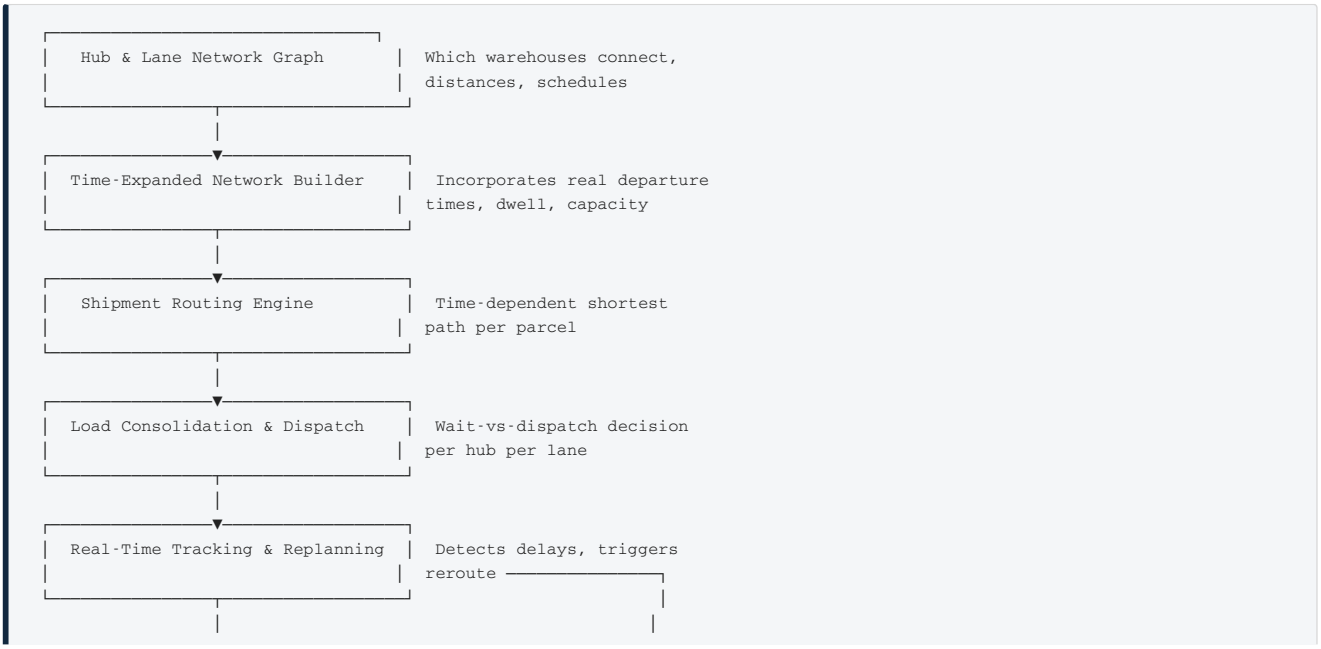
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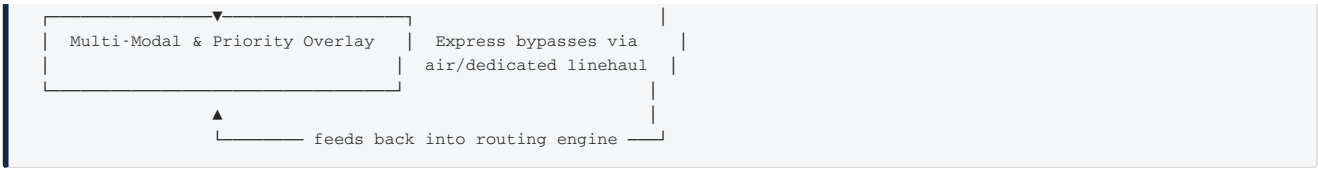
3. Questions to Ask If This Data Doesn't Exist Yet

- **"Do we have a defined lane network (which hub pairs run scheduled trucks), or is every route decided ad hoc?"** If ad hoc, this is the first thing to formalize — you can't do time-expanded routing without knowing what's actually scheduled.
- **"Do we have real-time visibility into truck location and remaining capacity, or only status updates at hub check-in/check-out?"** Without live tracking, dynamic re-routing (Section 7) can only react after a delay is already reported, not anticipate it.
- **"What are today's dispatch rules — fixed schedule regardless of load, or wait-until-full, or a hybrid?"** Many networks still run on ad hoc "whenever the truck feels full enough" judgment calls by depot staff — formalizing this (Section 6) is often the single highest-leverage improvement available.
- **"Is there a parcel volume forecast per lane, or is consolidation timing decided purely on what's physically at the dock right now?"** A forecast (even a simple one) changes "should I wait 30 more minutes" from a guess into a calculated decision.
- **"Are driver hours-of-service regulations already built into transit time estimates?"** A ~1,400 km Delhi–Mumbai route can't be legally driven non-stop by one driver — if this isn't already modeled, transit time estimates are likely optimistic.
- **"Do we operate any multi-modal capacity (rail, air cargo) or is this road-only?"** Changes the solution space substantially, especially for express-tier shipments.

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4. System Architecture: The Routing & Dispatch Pipeline





The strategic lane-network layer (top box) is redesigned periodically based on volume trends; everything below it runs continuously in near real time for each shipment.

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5. The Core Insight: Time-Expanded Network Routing

5.1 Why plain shortest-path-by-distance gets this wrong

A naive approach treats the hub network as a static graph and runs Dijkstra's algorithm to find the geographically/time-shortest chain of hubs. This fails in practice because it ignores the thing that actually determines when a parcel moves: **whether a truck is scheduled to leave that hub, on that lane, soon enough** — exactly the problem you flagged. A path that's shorter in kilometers can be *slower* in practice if it routes through a hub with only one truck departure per day that already left.

5.2 The right model: a time-expanded (time-space) network

Instead of nodes representing just *hubs*, nodes represent **(hub, time)** pairs. Edges are of two types:

- **Travel edges**: a scheduled truck departing Hub A at time `t1`, arriving Hub B at time `t2` (capacity-limited by truck size)
- **Waiting/dwell edges**: staying at the same hub from time `t` to `t + Δt`, representing a parcel sitting in queue for the next available departure or for consolidation

Finding the fastest route for a parcel becomes a **shortest path search over this time-expanded graph** — solved with a time-dependent Dijkstra or label-correcting algorithm — starting from (Delhi, order-ready-time) and ending at (Mumbai, deadline). This is structurally identical to how flight- or train-itinerary search engines work, and it's the correct formal answer to "how do I pick the next best warehouse": **you don't pick based on geographic proximity, you pick based on which (hub, time) node gets you to Mumbai soonest, given what's actually scheduled to depart from there.**

5.3 What this naturally solves

- If a geographically "obvious" next hub has no truck leaving for 10 hours, but a slightly less direct hub has one leaving in 30 minutes, the time-expanded shortest path will correctly prefer the second — no special-casing needed, it falls directly out of the algorithm.
- Hub processing/dwell time is included as part of the path cost, not ignored — so the model won't promise an unrealistic connection that ignores unloading/sorting/reloading time at each stop.

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6. Load Consolidation & the Dispatch Decision

You flagged this directly: sending a full truck for a single parcel isn't cost-effective, but waiting too long for a full load blows the delivery time. This is a classic **dispatch/consolidation policy** problem in freight operations, and there are three standard policy shapes:

Policy	Rule	Trade-off
Time-based (fixed schedule)	Truck departs at fixed times regardless of load (like a bus timetable)	Predictable ETA, but sometimes departs underfull (wasted cost), sometimes forces waiting even if full early (wasted time)
Quantity-based (fill threshold)	Truck departs once it reaches X% capacity	Cost-efficient, but ETA becomes unpredictable — depends entirely on how fast volume arrives
Hybrid (whichever comes first)	Depart at scheduled time OR when full, whichever happens first	Most commonly used in practice — bounds worst-case wait while still allowing early efficient departure

Formalizing the trade-off explicitly turns this from a judgment call into a calculation. Define a simple objective per potential dispatch decision:

```
Total cost = Transportation cost (≈ fixed cost per truck-km, largely load-independent)
           + λ × Expected delay cost (SLA penalty / customer dissatisfaction, scaled by shipment priority)
```

Where `λ` is tuned by shipment priority tier — **express shipments get a high λ** (dispatch fast even underfull, or route via a dedicated/faster path), **economy shipments get a low λ** (tolerate longer consolidation wait to reduce cost per parcel). This single parameter is what lets one network serve multiple SLA tiers without needing entirely separate routing logic.

A more advanced version uses a **short-horizon volume forecast** at the hub ("30 more parcels are due to arrive in the next 40 minutes based on inbound trucks already en route") to decide whether waiting is actually worth it — turning a static threshold into a forecast-informed dynamic decision.

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7. When No Truck Is Leaving on That Route: Dynamic Rerouting & Hub Tiering

This is really two related situations, both handled the same way once the time-expanded network exists:

7.1 No truck scheduled soon on the "obvious" next lane. The routing engine (Section 5) simply finds the next-best (hub, time) path — which may mean routing through a different intermediate hub than the geographically direct one, if that hub has a sooner departure toward the destination. This should be automatic, not a manual exception.

7.2 Hub hierarchy / feeder-to-consolidation design. Not every warehouse should be treated as equally capable of dispatching long-haul trucks. In practice, networks like this typically have a **tiered hub structure**:

- **Local/spoke warehouses** — smaller volume, may not justify their own long-haul trucks
- **Regional consolidation hubs** — higher volume, where several spoke warehouses' parcels get combined ("feeder" or shuttle runs from spokes into the regional hub), and from which full-truckload long-haul departures run on a real schedule
- **Major sortation hubs** — the largest nodes (this is how FedEx/UPS-style networks work, with a small number of "super-hub" sortation centers), where cross-country lanes actually originate

If a parcel arrives at a small spoke warehouse without enough volume to justify its own onward truck, the system's answer to "next best warehouse" is: **shuttle it via a smaller feeder vehicle to the nearest regional consolidation hub that does have a scheduled long-haul departure soon** — rather than waiting indefinitely at the spoke for enough local volume to accumulate. This is exactly the "figuring the next best warehouse" instinct you described, formalized.

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8. Factors You Didn't Mention — But That Matter

- **Driver hours-of-service regulations.** A ~1,400 km Delhi–Mumbai haul cannot legally be driven non-stop by a single driver. Real transit time must account for mandated rest breaks, and long lanes typically need either a **relay hub** (driver swap partway) or a **two-driver team truck** — this can add several hours that a naive distance/speed calculation would miss entirely.
- **Multi-modal options.** Indian Railways parcel/freight services and air cargo are viable alternatives to road for parts of this journey, especially for high-priority shipments — a mature system routes across modes, not just across road hubs, when it's faster or when road capacity is constrained.
- **Hub sorting/dock capacity, not just truck availability.** Even if a truck is scheduled and has room, the hub itself has a throughput limit (parcels sorted per hour) — during peak season a well-scheduled truck can still be delayed by a congested hub floor. This is a distinct constraint from "is there a truck," and needs its own capacity model.
- **Backhaul/empty-return optimization.** A truck running Delhi→Mumbai ideally carries a return load Mumbai→Delhi too — this doesn't change any single parcel's ETA, but it materially affects the network's overall cost structure and therefore how many lanes/trucks the company can afford to run, which indirectly affects service frequency.
- **Road/weather conditions specific to the corridor.** Monsoon flooding on certain stretches (e.g., parts of Gujarat/Maharashtra during monsoon season), highway construction, or known accident-prone stretches should feed into the time-expanded graph's edge times as dynamic adjustments, similar to real-time traffic layers in last-mile ETA systems.
- **Interstate regulatory friction.** E-way bill compliance, interstate check-posts, and RTO checks add real, somewhat predictable delay on Indian interstate freight — worth modeling as a fixed or semi-fixed time addition per state-border crossing rather than ignoring it.
- **Network resilience/contingency rerouting.** If a hub is disrupted (strike, accident, flooding) or a scheduled truck breaks down, the system needs an automatic fallback path — this is a natural extension of the time-expanded shortest-path model (simply remove the disrupted edge and re-solve) but needs to be triggered automatically from real-time signals, not manually.
- **Customer-facing live tracking and ETA updates.** The same real-time truck tracking that powers dynamic rerouting should also feed a customer-facing "where's my parcel" experience, recomputed at each hub transition — conceptually the same continuous re-estimation loop used in last-mile delivery ETA systems, just applied at inter-city hub granularity instead of GPS-ping granularity.

- **Fragile/hazmat/special-handling segregation.** Truck loading isn't pure bin-packing by volume/weight — certain parcel categories can't be co-loaded, or need specific placement (e.g., not at the bottom of a stack), which constrains the consolidation/loading optimization beyond simple capacity math.
- **Strategic network redesign feedback loop.** Actual transit time and delay data collected from the operational layer should periodically feed back into redesigning the strategic lane network itself (Section 1) — e.g., if a particular regional hub consistently struggles to fill trucks on schedule, that's a signal to adjust its lane frequency or reconsider its role in the hub hierarchy.

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9. Constraints That Shape This System

Constraint Type	Specifics
Lane availability	Not every hub pair has a direct scheduled truck — routing must respect the actual designed lane network, not an assumed full mesh
Truck capacity	Volume/weight limits per truck constrain both consolidation decisions and loading (bin-packing)
Driver hours-of-service	Legally mandated rest periods cap how far a single driver/truck can go without a relay or team-driving arrangement
Hub processing capacity	Sorting/dock throughput at each hub is a separate bottleneck from truck/lane availability
Cost vs. delay trade-off	Dispatching underfull trucks is costly; waiting too long breaches SLA — must be explicitly balanced, ideally per shipment-priority tier
Road/weather conditions	Seasonal and route-specific risk (monsoon flooding, highway construction) affects real transit time beyond static distance/speed assumptions
Regulatory/compliance	Interstate e-way bills, check-post procedures, and RTO checks add real (if roughly predictable) delay at state borders
Multi-modal integration	If rail/air options exist, the routing engine needs to compare across modes, not just across road hubs
Network resilience	The system needs a fallback path when a hub or lane is disrupted, not just an optimal path under ideal conditions
Backhaul economics	While not a per-parcel constraint, empty-return inefficiency affects how many lanes/trucks the network can sustain, indirectly capping service frequency
Real-time visibility	Dynamic rerouting is only as good as the freshness of truck location and hub status data feeding it

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10. Base Assumptions

1. The company already operates (or can define) a fixed-ish lane network between hubs rather than deciding every route from scratch — full ad hoc routing with zero defined lanes would be a much bigger foundational build than assumed here.
2. Road is the primary mode for this corridor, with rail/air as optional overlays for high-priority shipments rather than the default.
3. Hubs are tiered (spoke vs. regional consolidation vs. major sortation) rather than uniformly capable — if every warehouse is truly identical in role and capacity, the hub-hierarchy recommendation in Section 7 simplifies but the time-expanded routing logic still applies.
4. There's at least basic real-time truck tracking today (even if not yet used for dynamic re-planning) — building GPS tracking from zero would be a larger separate initiative.
5. Shipments have some notion of priority/SLA tier (even informally) — if every parcel is treated identically regardless of urgency, the cost-delay trade-off in Section 6 still works but with a single shared λ rather than tier-specific values.

Document prepared as a system design framework for inter-city middle-mile parcel routing, load consolidation, and dispatch decisions in a hub-and-spoke logistics network.